Mapping arguments to key point: Match Scoring of arguments using sentence embedding and MoverScore without labelled data

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Abstract There might be hundreds of arguments for a single debate topic. and it is usually difficult to read them all. Therefore, some of the existing studies have attempted to summarize and aggregate these arguments manually into about 14 types of content (key point) in order to make them easier to understand. Some researchers pair key points with arguments and use BERT to identify whether an argument and a key point are matching or not. However, the previous study averaged word vectors to compute sentence vectors, which resulted in the unbalance of important word information. This means that the important words should be weighted more than other words, but the previous study does not consider about that. To solve this problem, we implemented a sentence embedding model called Sentence-BERT, which is fine-tuned on NLI dataset with BERT. The results are better than the existing SOTA method, i.e. BERT. Furthermore, to compare each words between two sentences, we introduce MoverScore, which yield the best results.

Keywords Natural Language Processing, Argument Mining, Sentence-BERT, MoverScore, Debates

1 Introduction

Arguments in debates or discussions are broadly studied in NLP area [1, 2]. Research field related to arguments is called argument mining, and this field has been increasingly famous in this ten years. In argument mining, we basically extract the argument structures from unstructured texts or utterances [3]. These attempts for detecting argument components (e.g. claims, premises) have been clarifying where an author (or a speaker) mainly claims, where is an evidence for their claim, and where they show an example for their argument. Nowadays, researchers are getting better at distinguishing above-mentioned components, therefore targets of the area of argument mining are gradually moved on to different topics. Argument mining researchers have focused on predicting persuasiveness of an argument [4, 5], filling gaps between claims [6, 7], and predicting argument stance (pro/con) [8, 9]. These works have benefited from enormous amount of argument data from debates, discussions, speeches, or essays.

However, studies on aggregating such numerous arguments seem to be neglected. Numerous arguments make it hard to know which argument is better to read. Although there are various and numerous studies in the field of argument mining, no method will be practical if users cannot find accurate opinions for their debating topic. We have to narrow down the target argument, i.e. summarize a lot of arguments into one core argument. To address this problem, we also need plentiful arguments, and a new question arises; how to gather arguments? There are typically two approaches to aggregating arguments; one way is to retrieve them from online debate resources (like *iDebate* or *CreateDebate*), another is to gather arguments from debate spectators who express their opinions [2]. Using both methods, the number of gathered arguments in a single debate topic often exceed hundreds or more. To avoid laborious trouble, reading through such amounts of text, one need to work on summarizing arguments.

In order to address this problem, Reimers et al. [10] use capabilities of contextualized word embeddings of ELMo [11] and BERT [12] to classify and cluster topic-dependent arguments from Argument Facet Similarity Corpus [13]. They clustered related arguments, however they did not generate summaries of arguments. This problem is mentioned in Bar-Haim et al. [14], who defined the key point: usually single sentence describing a set of similar arguments, i.e. a summary of arguments. A large dataset consisting of [argument, key point] pairs is generated. Next, two experimental steps are conducted: *Match Scoring* and *Match Classification*. In *Match Scoring*, a match score for a given [argument, key point] pair is computed, (see Table 1, which

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Topic: **Homeschooling should be banned**

Arguments	Key Points	Stance	Match
education at home could represent a risk since it is not regulated	Homeschools cannot be regulated/standardized	1	1
a parent knows best what is good for their kid	Parents should be permitted to choose the education of their children	-1	0

Table 1: Example of [argument, key point] pairs for a given topic. Stance and Match refer to pro (1) or con (-1), and the same meaning (1) or not (0), respectively.

shows examples of the dataset for *Match Scoring*, named ArgKP). In *Match Classification*, their methods discover matching key points for each argument. For these tasks, they describe that BERT-large [12] fine-tuned model is achieving the best results in supervised learning, and a BERT-large embedding method is the best among unsupervised methods. Although Bar-Haim et al. claim that their research utilizes argument summarization, only the relevance between key points and arguments is calculated, which means their both experiments are almost identical to the above-mentioned research of Reimers et al. [10]. Furthermore, for *Match Scoring* and *Match Classification*, creating key points is necessary beforehand. This indicates that their methods cannot be used with new debate topics for which key points do not exist.

The mapping step is useful when generating key point from a set of arguments. *Match Scoring*, which is explained above, judges whether a given argument and a given key point are related or not. Following this *Match Scoring* step, it is possible to create a set of arguments for generating a key point.

In this paper, we address the *Match Scoring* task for key point generation with the ArgKP dataset [14], which is explained in Section 3.2. We assume that the important factor for summarizing large number of arguments into one key point is to map an argument to key points, not to select a key point from one argument. Therefore, we adopt the *Match Scoring*, not the *Match Classification*, for our target task in this paper. The previous research [14] experimented with both supervised and unsupervised methods, however we tackle the task with unsupervised approach because the size of the data is too small when we utilize the ArgKP dataset for key point generation. The previous unsupervised approaches only averaged word vectors for computing a sentence vector, so that the weight of important word information is identical with other words during sentence vector calculation. To address this problem, we try to solve it applying two methods: Sentence-BERT [15] and MoverScore [16]. We hypothesize that sentence embedding models could be a good solution to represent a whole sentence in a vector, and that output vector will not be able to represent significant words in an argument. Therefore, we utilize variations of Sentence-BERT model, which is fine-tuned with NLI dataset and is able to represent sentence embeddings. We also utilize MoverScore, a metric to measure text similarity using Word Mover's Distance [17] in order to calculate similarity between an argument and a key point on the word level. To the best of authors' knowledge, this is the first attempt to address the important word problem with sentence embedding models and MoverScore metric for the *Match Scoring* task.

The main contributions of this article are:

1. Pointing out that existing methods which average word embeddings to compute sentence embeddings, cannot weight important word words in an argument;
2. Applying two methods to put more weight on significant words and less on non-significant words, i.e. Sentence-BERT and MoverScore;
3. Experimentally proving that both Sentence-BERT and MoverScore methods outperform the existing unsupervised methods, and the MoverScore approach yield the best result.

2 Related Works

This section describes works related to this study. First section explains the history of the field, and similar topics in the area of argument mining. The second section presents approaches to measure text similarity.

2.1 Argument Mining

Argument mining has started with researchers working on argument components [3] for debates, discussions or student essays. After argument structure had been gradually

solved, several researchers began to work on argument persuasiveness [4] or persuasiveness of the whole debate [18] in order to generate more convincing arguments. Stance prediction is also active research topic on argument mining. Several works estimate whether a given argument is following a given debate topic or not (pro/con) [8, 9]. In order to conduct research on arguments from several perspectives, various large argument datasets have been recently created [19, 20]. Stab et al. [19] also focus on retrieving arguments from different sources. Their dataset covers about 90% of arguments found in expert-curated lists of arguments from an online debate portal. Similarly, ArgKP dataset conducted by Bar-Haim et al. [14] also indicates high agreement between expert dataset (key point list) and arguments, achieving Cohen's kappa=0.82.

More recently, Misra et al. [13] attempt to measure a similarity of arguments in debate portal. They provide Argument Facet Similarity Corpus. The purpose of their work is to identify whether an argument has the same facets as other arguments in across multiple conversations. Several researchers are following their topic with logical rules or contextual embedding models [21, 22]. These works are similar to mapping arguments to key points, but they do not calculate similarity between a facet and an argument. Moreover, they do not clarify key points, and in this aspect, their work clearly differs from ours.

As for mapping arguments to key points, i.e. *Match Scoring* and *Match Classification* tasks, existing works [23, 14] can be given as most known examples. Boltužić and Šnajder [23] implement argument clustering. They map arguments derived from one online debate portal (*ProCon*) to arguments of other portal (*iDebate*). However, they create only two debate topics with less than 400 arguments. Bar-Haim et al. [14] make a massive dataset including pairs of an argument and a key point labelled with relevancy. They also propose *Match Scoring* and *Match Classification* as the first steps to summarize arguments.

2.2 Text Similarity

NLP researchers recently utilize vectors of words / documents for measuring similarity between texts. Today, we usually compute word vectors with word embedding, sentence embedding, or language models. One of the most well-known algorithms, word2vec [24] is a pre-trained skip-gram or Continuous Bag-of-Words (CBOW) model to represent words as vectors. Following the pre-trained word embedding models, sentence / document embedding models have been also developed. For example, Skip-Thought Vector [25] is based on a sentence encoder that predicts the surrounding sentences of a given sentence. Skip-Thought is based on an encoder-decoder model, its encoder maps words to a sentence vector and its decoder generates the surrounding sentences. Sentence Transformers [15] are sentence embedding models for English texts based on siamese / triplet networks [26, 27]. They insert language models (like BERT [12]) into the network of siamese / triplet networks. They call BERT-based siamese / triplet networks as "Sentence-BERT."

There are also some other methods to compute text similarity. Word Mover's Distance [17] algorithm calculates the distance between texts with word2vec, which introduces Earth Mover's Distance [28] into NLP field. Inspired by Word Mover's Distance, Zhao et al. [16] investigate encoding systems to devise a metric that shows a high correlation with human judgment of text quality in summarization or machine translation tasks. They name their metric MoverScore, whose details are described in Section 5.3.

Our proposed methods are based on Sentence-BERT and MoverScore, which are described above.

3 Experimental Setup

In this section, we describe the preparation for *Match Scoring* task; the task setting, and the data for our experiments.

3.1 Match Scoring

In *Match Scoring*, we map arguments to a key point which may have similar meaning to them, and compute a match score for [argument, key point] pairs. As examples of matching and not matching pairs, in Table 1, we show pairs, their stance to a given topic, and the manually annotated matching label. We try to identify exactly whether such arguments and key points are similar or not. We evaluate match scores with accuracy, precision, recall, and F1.

3.2 Data

We utilize ArgKP dataset [14] for *Match Scoring*. This dataset consists of about 24,000 [argument, key point] pairs which are labeled as follows; (1) whether an argument and a key point describe the same content or not; and (2) whether an argument represents pro side or con side. To the best of authors' knowledge, ArgKP is the largest dataset usable for the *Match Scoring*. It is based on IBM-Rank-30k dataset [29] which contains 30,497 arguments annotated for their quality. Using the quality-indicating labels, Bar-Haim et al. [14] filter out unclear arguments whose quality score is

lower than 0.5 or polarity score is lower than 0.6. After the arguments are filtered, they generate key points, and pair an argument with a key point. They annotate a [argument, key point] pair with the label whether they are similar to each other or not. For each stance in one topic, 6.75 key points were generated on average, and there are 378 key points in total. With 24,000 [argument, key point] pairs in ArgKP dataset, we divide the dataset into 7 test topics and 21 train topics, following experimental setup of the previous research [14].

4 Baseline Methods

For *Match Scoring* task, we adopt BERT-based and Word2Vec with WMD methods for comparing with our proposed approaches described in the Section 5.

4.1 BERT-large Fine-tuned

Bar-Haim et al. experimentally showed that fine-tuned BERT-large model yielded the best score in the several supervised methods [14]. Note that their approach is supervised, unlike the others.

4.2 BERT-large Embedding

For comparison with other unsupervised approaches, we choose BERT-large [12] as the SOTA system for *Match Scoring* [14]. When we embed an argument and a key point using BERT-large, we examine averaged word embeddings as their embeddings. If the cosine similarity between an argument and a key point is above a threshold decided with the train data, the methods regard the pair as relevant.

4.3 Word2Vec without WMD

When we embed an argument and a key point using Word2Vec, we examine averaged word embeddings as sentence embeddings. If the cosine similarity between an argument and a key point is above a threshold decided with the train data, the methods regard the pair as relevant.

4.4 Word2Vec with WMD

We experiment with Word Mover's Distance (WMD) [17] to compare with MoverScore-based method. We compute distance between an argument and a key point with WMD. If the dissimilarity between an argument and a key point is BELOW a threshold decided with the train data, the methods regard the pair as relevant.

5 Proposed Methods

This section describes our proposed methods for *Match Scoring* task, i.e. Sentence-BERT, Sentence-BERT fine-tuned on STS benchmarks, and MoverScore approaches. All thresholds for similarity function are acquired from the train data for each method. The detailed way to decide thresholds is following;

1. Calculating F1 score for the positive (matching) class for each threshold by clustering the train data;
2. Obtaining the thresholds which maximize the F1 score for the train data.

As a result, the thresholds of Sentence-BERT-base, Sentence-BERT-large, Sentence-BERT-STSb-base, Sentence-BERT-STSb-large, and MoverScore are 0.64, 0.71, 0.58, 0.56, and 0.12 respectively.

5.1 SBERT

Siamese Neural Networks [30] consist of two neural networks trained with the same weights. These weights are trained on two different input vectors. Reimers and Gurevych [15] proposed Siamese BERT Network and implemented it as Sentence-BERT[1] (SBERT for short) trained on Natural Language Inference (NLI), SNLI and MultiNLI datasets, in order to create universal sentence embeddings. Following the original Sentence-BERT, we use the mean-pooling model.

5.2 SBERT-STSb

SBERT-STSb model is also proposed in [15]. These models are first fine-tuned on the AllNLI dataset, then on the train set of STS benchmark. With this fine-tuning phrase of training process, SBERT-STSb is well suited for measuring semantic textual similarity.

5.3 MoverScore

MoverScore [16] is a robust evaluation metric to evaluate summarization or machine translation tasks. We adopt this scoring metric to calculate similarity between an argument and a key point for calculating distances between words in each sentence. MoverScore is based on WMD or Sentence Mover's Distance (SMD) [31]. To measure semantic distance, n-gram (n=1, 2) is utilized in addition to word distances. For embedding model calculating word distances, Word2Vec, ELMo, and BERT-base are adopted. BERT-base is fine-tuned on one among following datasets: MNLI, QANLI, or QQP. Furthermore, the word representations from ELMo and BERT-base are aggregated with Power Means or Routing Mechanism. ELMo and BERT output different vectors from each layer, therefore they aggregate vectors (all three layers for ELMo, and the final five layers for BERT-base).

[1] We use Sentence-BERT model available at https://github.com/UKPLab/sentence-transformers

	Acc	P	R	F1
BERT-large Fine-tuned [14]	0.868	0.685	0.688	0.684
BERT-large Embedding [14]	0.660	0.319	0.550	0.403
Word2Vec w/o WMD	0.670	0.290	0.501	0.368
Word2Vec w/ WMD	0.493	0.218	**0.635**	0.324
SBERT-base	0.682	0.317	0.571	0.408
SBERT-large	0.728	0.354	0.513	0.419
SBERT-STSb-base	0.713	0.341	0.538	0.418
SBERT-STSb-large	0.706	0.335	0.544	0.415
MoverScore	**0.747**	**0.388**	0.554	**0.457**

Table 2: Results of each clustering embedding method. Acc, P, and R refer to accuracy, precision, and recall, respectively.

We use the best metric for summarization tasks given in [16], i.e. uni-gram, BERT-base, MNLI, Power Means, and WMD, for our proposed method.

6 Evaluation Results

The results of accuracy, precision, recall, and F1 score on the test dataset are shown in Table 2. These are results of experiments with the test data using thresholds learned from the train data. As shown in Table 2, SBERT-large achieves higher scores in accuracy, precision, and recall than existing unsupervised SOTA, i.e. BERT Embedding. While SBERT-large achieves a relatively good score, versions of SBERT-STSb do not exceed SBERT-large even though they have been reported as having superior capability for measuring semantic similarity [15]. For this reason, before the experiment, we assumed that SBERT-STSb would exceed SBERT score. STS benchmark and other NLI datasets contain daily conversation-like data. Our intuition is that such data might be unsuitable for debate topics. In future, debate or argument-oriented BERT model could be used instead of the standard BERT [20].

In addition to SBERT, MoverScore yields the best scores except recall. MoverScore relies on soft-alignment (many-to-one), so it allows to map semantically related words in one text to the respective word in another text. This is, in our opinion, why the MoverScore performed the best. Regarding recall score, the problem seems to lay in the dataset bias. As we have already mentioned in Section 3.2, more non-matching data exist than matching ones. This is most probably the reason why the recall value in Word2Vec with WMD is higher than that in MoverScore.

7 Error Analysis

Within the topic "We should ban the use of child actors," argument "the use of child actors exploits a child and can have negative effects on them" and key point "Being a performer harms the child's education" represent different content. However, the similarity between them is high (0.9140). The reason for this may be that both of sentences describe negative effects on child actors, so the method incorrectly classified them as representing the same content. Both argue the negative effects, but from different facets. It can be assumed that our model will obtain better results if it can identify the difference from a perspective of detecting a different facet of a discussed problem. This problem occurs in both of our methods, SBERT and MoverScore. We plan to treat it in near future.

8 Conclusion

Our work is focused on the existing task, *Match Scoring* for the first step toward key point generation. We point out that the existing methods, averaging word embeddings to compute sentence embeddings, cannot weight important words in an argument, therefore we propose to utilize Sentence-BERT model and MoverScore metric for the task, and both of our approaches outperformed the state of the art system (BERT-large) on the ArgKP dataset [14]. This could be due to the fact that our methods weight important words which is not done in other studies. From these results, we can say the sentence embedding models and NLI dataset, which is used for fine-tuning / training Sentence-BERT and MoverScore metric, are useful for the *Match Scoring* task, so we plan to utilize other sentence transformers, like Sentence-RoBERTa which is also fine-tuned on NLI dataset, to improve our results. MoverScore is based on Word Mover's Distance, which means word significance treats A and word meaning as the same vactor. To solve this problem, we are going to introduce Word Rotator's Distance, which can represent the importance and the meaning separately.

For the next step, we will attempt to generate key points from argument set as mentioned in Section 1. We plan to use Maximal Marginal Relevance (MMR), which can identify important words in texts, for summarizing arguments in order to generate key points.

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